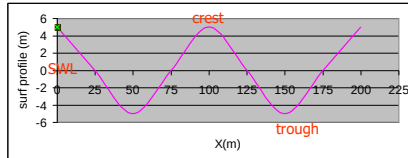


Wave profile and definitions



- **SWL** (Still Water Level), water surface when there is no wave action
- **Wave profile**(η), vertical displacement of water surface
- **Wave crest**, the highest point
- **Wave trough**, the lowest point
- **Wave amplitude** (a), distance betw. SWL and the crest
- **Wave height** (H) = $2a$
- **Wave length** (L), hor. Dist betw two succeeding crest
- **Wave period** (T), time int. betw two succeeding crest

WAVE PERIOD, T



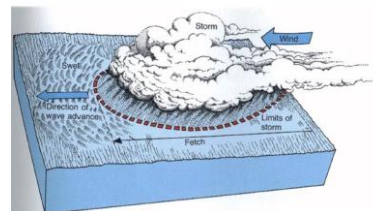
WAVE CLASSIFICATION

Ocean waves can be classified in various ways. One classification uses the forces which generate the waves:

- **Meteorological forcing** (wind, air pressure); sea and swell belong to this category.
- **Earthquakes and land slides**; they generate tsunamis, which are long waves.
- **Tides** (astronomical forcing); they are long waves as well

WIND WAVE GENERATION

- Waves are generated by the tangential (shear) and the normal (pressure) stresses exerted by the wind over the sea surface



Water Wave - A deformation of the water's surface.

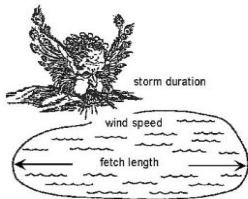


Figure from: http://geology.spm.edu/Reichert2_image/wavegen.gif

Duration - length of time the wind blows.

Wind speed - How fast the wind is blowing.

Fetch - The horizontal distance (in direction of the wind) over which a wind blows.



BEAUFORT FORCE 1
WIND SPEED: 1-3 KNOTS

SEA: WAVE HEIGHT .1M (.25FT), RIPPLES WITH THE APPEARANCE OF SCALES, BUT WITHOUT FOAM CRESTS



BEAUFORT FORCE 4
WIND SPEED: 11-16 KNOTS

SEA: WAVE HEIGHT 1-1.5M (3.5-5FT), SMALL WAVES BECOMING LONGER, FAIRLY FREQUENT WHITE HORSES



BEAUFORT FORCE 7
WIND SPEED: 28-33 KNOTS

SEA: WAVE HEIGHT 4-5.5M (13.5-19 FT), SEA HEAPS UP, WHITE FOAM FROM BREAKING WAVES BEGINS TO BE BLOWN IN STREAKS ALONG THE WIND DIRECTION



BEAUFORT FORCE 10
WIND SPEED: 48-55 KNOTS

SEA: WAVE HEIGHT 9-12.5M (29-41FT). VERY HIGH WAVES WITH LONG OVERHANGING CRESTS, THE RESULTING FOAM, IN GREAT PATCHES, IS BLOWN IN DENSE WHITE STREAKS ALONG WIND DIRECTION. ON THE WHOLE, SEA SURFACE TAKES A WHITE APPEARANCE. TUMBLING OF THE SEA IS HEAVY AND SHOCK-LIKE, VISIBILITY AFFECTED.



BEAUFORT FORCE 12
WIND SPEED: 64 KNOTS

SEA: SEA COMPLETELY WHITE WITH DRIVING SPRAY, VISIBILITY VERY SERIOUSLY AFFECTED. THE AIR IS FILLED WITH FOAM AND SPRAY

OTHER WAVES

- SWELL (Soluğan, ölü deniz dalgası)
- Waves generated by another storm
 $15s < T < 30s$
 More orderly than wind waves (like regular)
 $S_{max} > 25$ (wind waves=10)

swell

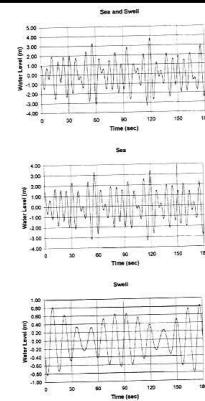


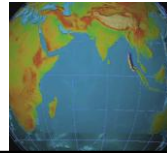
Figure 2.5 Sea and Swell Separated

The Great Wave of Kanagawa
(by Katsushika Hokusai (1760 – 1849))

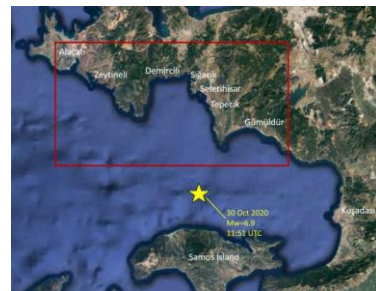


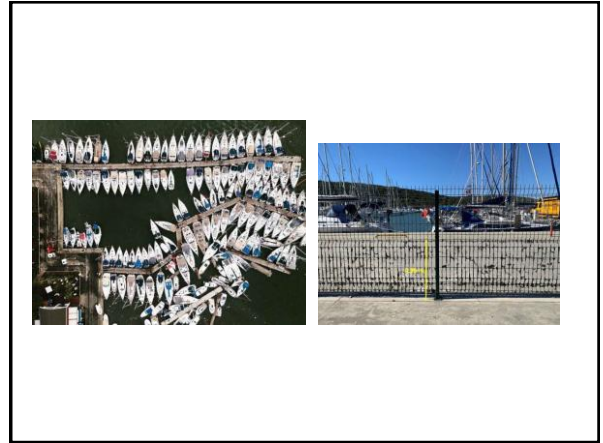
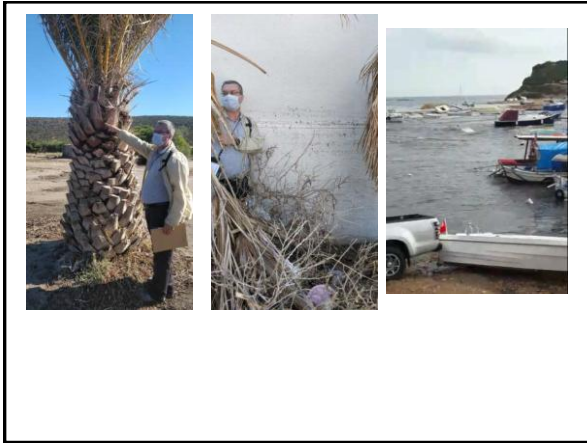
TSUNAMI

- Caused by earthquake, landslides, volcanic eruptions etc.
- $T=5-60$ min
- $H=0.5-1$ m in deep water (high in SW)
- Celerity depends on depth ($4500\text{m}=756\text{km/hr}$)



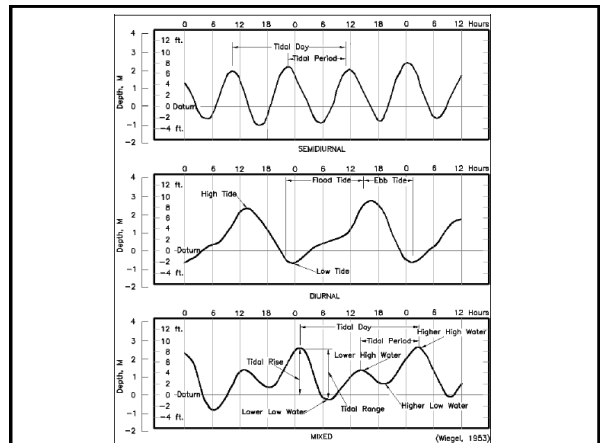
Translatory motion not oscillatory





TIDAL WAVE

- The gravitational attraction between earth, moon and sun
- The centrifugal force resulting from the relative motion of the earth, moon and sun
- $T=12-24$ hrs (tidal period)
- High water level-low water level=tidal range= up to 10m, in Turkey= 30 cm
- Important in the design



WATER WAVE MECHANICS

wave theories

Eqn of motion and continuity

for a frictionless (ideal) fluid

irrotational motion (motion is free of vortices)

incompressible (density is constant)

surface tension is negligible

under appropriate boundary conditions =Stokes wave theory

H/L , H/d small, neglect nonlinear terms=Small amplitude wave theory, Airy

Cnoidal wave theory shallow water

Solitary wave theory very shallow water

Laplace
Bernolli

Wave profile

$$\eta = a \cos(kx - \sigma t) = a \cos\left(\frac{2\pi}{L}x - \frac{2\pi}{T}t\right)$$

- Wave frequency (f)
- Wave number (k)
- Angular wave frequency(σ , ω)
- Wave celerity (C)

Basic wave parameters

- Wave height (amplitude)
- Wave length (period)
- Water depth

Relation between L and d

$$L = \frac{gT^2}{2\pi} \tanh\left(\frac{2\pi d}{L}\right)$$

$$\sinh(x) = 0.5(e^x - e^{-x})$$

$$\cosh(x) = 0.5(e^x + e^{-x})$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{As } \frac{2\pi d}{L} \rightarrow 0 \rightarrow L = gT^2/2\pi = 1.56T^2 = L_0 \rightarrow L = L_0 \tanh\left(\frac{2\pi d}{L}\right)$$

For deep water

$$L = L_0 \tanh\left(\frac{2\pi d}{L}\right)$$

Eckart (1952) gives an approximate expression for the above implicit equation

$$L \cong L_0 \sqrt{\tanh\left(\frac{2\pi d}{L_0}\right)}$$

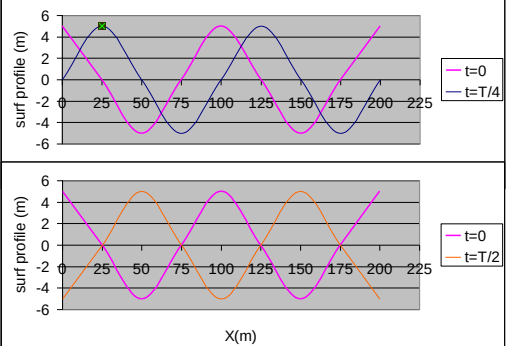
Table 4.3 Gravity Wave Table (Lyngby, 1974)

GRAVITY WAVE TABLE									
L	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$	$\frac{2\pi d}{L}$
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.01	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.02	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.03	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.04	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.05	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.06	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.07	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.08	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.09	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.10	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.11	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.13	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.14	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.15	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.16	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.17	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.18	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.19	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.20	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.21	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.22	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.23	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.24	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.25	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.26	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.27	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.28	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.29	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.30	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.31	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.32	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.33	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.34	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.35	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.36	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.37	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.38	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.39	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.40	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.41	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.42	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.43	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.44	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.45	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.46	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.47	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.48	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.49	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.50	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

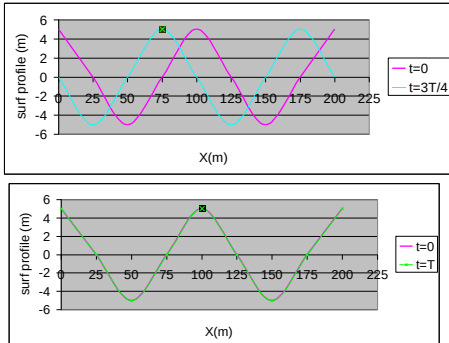
Example question

- $T=8\text{sec}$
- $d=10\text{m}$
- $L=?$
- $d/L=?$
- Plot wave profile at $t=0$

Wave propagation



Wave propagation



$$V = dx/dt$$

$$C = L/T$$

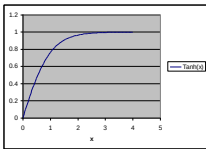
Wave celerity

$$L = \frac{gT^2}{2\pi} \tanh\left(\frac{2\pi d}{L}\right)$$

$$C = \frac{gT}{2\pi} \tanh\left(\frac{2\pi d}{L}\right)$$

Dispersion equation:
Longer period waves travel faster than
shorter period waves

Limits of Deep water and shallow water



$$\sinh(x) = 0.5(e^x - e^{-x})$$

$$\cosh(x) = 0.5(e^x + e^{-x})$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$x \rightarrow 0 \quad e^{-x} \rightarrow 0$$

$$\tanh(x) \rightarrow 1$$

$$\sinh(x) \rightarrow 0.5e^x$$

$$\cosh(x) \rightarrow 0.5e^x$$

DEEP WATER

$x = ?$ For DW

$$L = L_0 \tanh\left(\frac{2\pi d}{L}\right)$$

$$x = \frac{2\pi d}{L} > \pi$$

$$\frac{d}{L} = \frac{d}{L_0} > \frac{1}{2}$$

Limits of Deep water and shallow water

$$\sinh(x) = 0.5(e^x - e^{-x})$$

$$\cosh(x) = 0.5(e^x + e^{-x})$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$x \rightarrow 0 \rightarrow$$

$$\sinh(kd) = kd$$

$$\cosh(kd) = 1$$

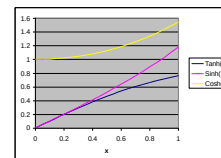
$$\tanh(kd) = kd$$

SHALLOW WATER

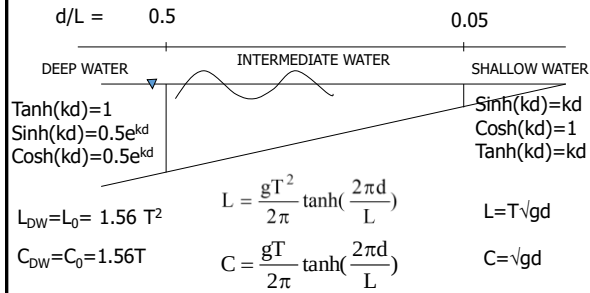
$x = ?$ For SW

$$x = \frac{2\pi d}{L} < \frac{\pi}{10}$$

$$\frac{d}{L} < \frac{1}{20}$$



Limits of Deep Water and Shallow Water



Example (CEM)

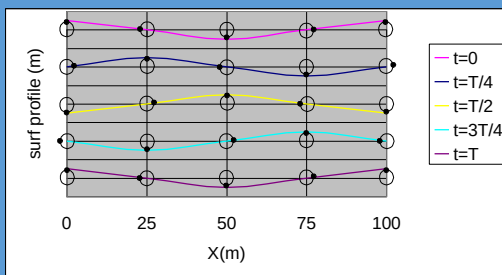
FIND:

The wave celerities C and lengths L corresponding to depths $d = 200$ meters (656 ft) and $d = 3$ m (9.8 ft).

GIVEN:

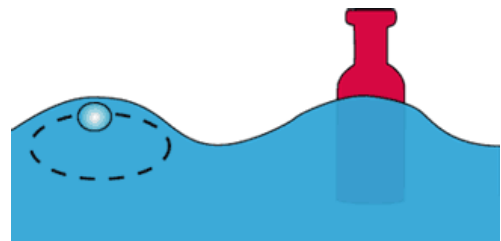
A wave with a period $T = 10$ seconds is propagated shoreward over a uniformly sloping shelf from a depth $d = 200$ m (656 ft) to a depth $d = 3$ m (9.8 ft).

WATER PARTICLE MOTION

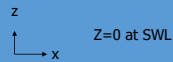


Waves propagate with velocity C but the individual water particles do not propagate. They move in particle orbits.

FACT - Deep water particles do not move forward with water wave, but instead move in elliptical orbits. Waves are the forward motion of energy, not water!



Water particle velocities

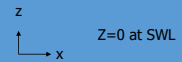


$$u = \frac{\pi H}{T} \frac{\cosh(k(z+d))}{\sinh(kd)} \cos(kx - \sigma t)$$

$$w = \frac{\pi H}{T} \frac{\sinh(k(z+d))}{\sinh(kd)} \sin(kx - \sigma t)$$

Water particle velocities

DEEP WATER



$$u = \frac{\pi H}{T} \frac{\cosh(k(z+d))}{\sinh(kd)} \cos(kx - \sigma t)$$

$$u_{DW} = \frac{\pi H}{T} e^{kz} \cos(kx - \sigma t)$$

$$\begin{aligned} \tanh(kd) &= 1 \\ \sinh(kd) &= 0.5e^{kd} \\ \cosh(kd) &= 0.5e^{kd} \end{aligned}$$

At $z=0$ (SWL)

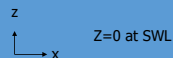
$$u_{DW, z=0} = \frac{\pi H}{T} \cos(kx - \sigma t)$$

At $z=-d$ (sea bottom)

$$u_{DW, z=-d} = 0$$

Water particle velocities

DEEP WATER



$$w = \frac{\pi H}{T} \frac{\sinh(k(z+d))}{\sinh(kd)} \sin(kx - \sigma t)$$

$$w_{DW} = \frac{\pi H}{T} e^{kz} \sin(kx - \sigma t)$$

$$\begin{aligned} \tanh(kd) &= 1 \\ \sinh(kd) &= 0.5e^{kd} \\ \cosh(kd) &= 0.5e^{kd} \end{aligned}$$

At $z=0$ (SWL)

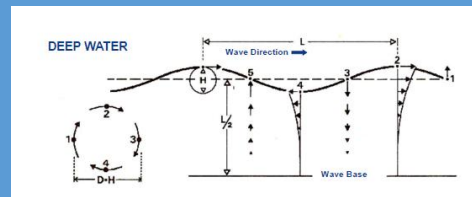
$$w_{DW, z=0} = \frac{\pi H}{T} \sin(kx - \sigma t)$$

At $z=-d$ (sea bottom)

$$w_{DW, z=-d} = 0$$

Water particle velocities

- Deep water
- $d/L = d/L_0 > 0.5$
- $u=w=0$ at $z=-d$



Water particle velocities

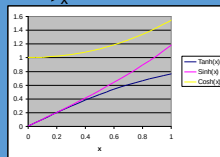
SHALLOW WATER

$$u = \frac{\pi H}{T} \frac{\cosh(k(z+d))}{\sinh(kd)} \cos(kx - \sigma t)$$

$$u_{SW} = \frac{\pi H}{T} \frac{1}{kd} \cos(kx - \sigma t)$$

At $z=0$ (SWL)

$$u_{SW,z=0} = \frac{\pi H}{T} \frac{1}{kd} \cos(kx - \sigma t)$$

 z
 x
 $Z=0$ at SWL
At $z=-d$ (sea bottom)

$$u_{SW,z=-d} = \frac{\pi H}{T} \frac{1}{kd} \cos(kx - \sigma t)$$

Water particle velocities

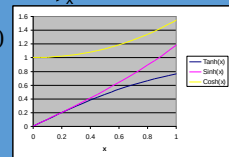
SHALLOW WATER

$$w = \frac{\pi H}{T} \frac{\sinh(k(z+d))}{\sinh(kd)} \sin(kx - \sigma t)$$

$$w_{SW} = \frac{\pi H}{T} \left(1 + \frac{z}{d}\right) \sin(kx - \sigma t)$$

At $z=0$ (SWL)

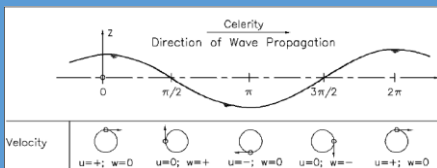
$$w_{SW,z=0} = \frac{\pi H}{T} \sin(kx - \sigma t)$$

 z
 x
 $Z=0$ at SWL
At $z=-d$ (sea bottom)

$$w_{SW,z=-d} = 0$$

Water particle velocities

- $w=0$ at $z=-d$
- U is max at crest and trough where $w=0$
- At $x=L/4, 3L/4, 5L/4, \dots$ w is maximum where $u=0$
- Therefore periodic variation in X direction
- u & w are out of phase (by an angle $\pi/2$)



Water particle velocities

 $d/L = 0.5$ 0.05

DEEP WATER

INTERMEDIATE WATER

SHALLOW WATER

$$u_{DW} = \frac{\pi H}{T} e^{kz} \cos(kx - \sigma t)$$

$$u = \frac{\pi H}{T} \frac{\cosh(k(z+d))}{\sinh(kd)} \cos(kx - \sigma t)$$

$$u_{SW} = \frac{\pi H}{T} \frac{1}{kd} \cos(kx - \sigma t)$$

$$w_{DW} = \frac{\pi H}{T} e^{kz} \sin(kx - \sigma t)$$

$$w = \frac{\pi H}{T} \frac{\sinh(k(z+d))}{\sinh(kd)} \sin(kx - \sigma t)$$

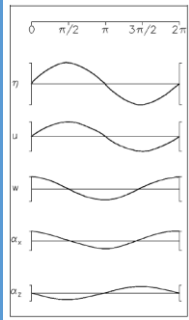
$$w_{SW} = \frac{\pi H}{T} \left(1 + \frac{z}{d}\right) \sin(kx - \sigma t)$$

 $w_{z=-d} = 0$ (always)

$$u_{DW,z=-d} = 0$$

- Abs(u) is max. at crest and trough where $w=0$
- Abs(w) is max. At $x=L/4, 3L/4, 5L/4, \dots$ where $u=0$
- Exponential decay in Z direction

Water particle velocities



Particle Trajectory



$$\xi = \int u dt = -a \frac{\cosh(k(z+d))}{\sinh(kd)} \sin(kx - \sigma t) \dots (1)$$

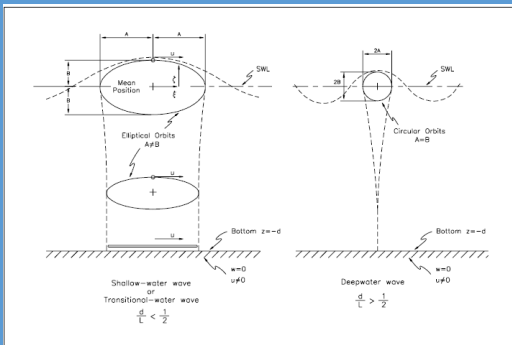
$$\epsilon = \int w dt = a \frac{\sinh(k(z+d))}{\sinh(kd)} \cos(kx - \sigma t) \dots (2)$$

From Eq 1..... $\sin^2(kx - \sigma t) = \left(\frac{\xi}{a} \frac{\sinh(kd)}{\cosh(k(z+d))} \right)^2$

From Eq 2..... $\cos^2(kx - \sigma t) = \left(\frac{\epsilon}{a} \frac{\sinh(kd)}{\sinh(k(z+d))} \right)^2$

$$1 = \frac{\xi^2}{A^2} + \frac{\epsilon^2}{B^2} \quad \text{Eqn of ellipse}$$

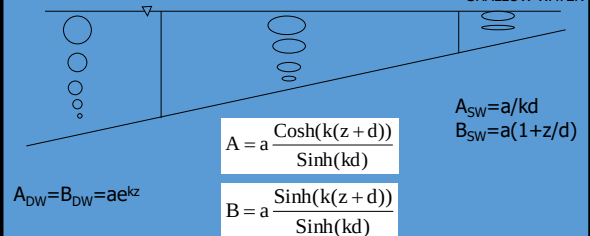
Particle Trajectory



Particle Trajectory

$d/L =$ 0.5 0.05

DEEP WATER INTERMEDIATE WATER SHALLOW WATER



$$A = a \frac{\cosh(k(z+d))}{\sinh(kd)}$$

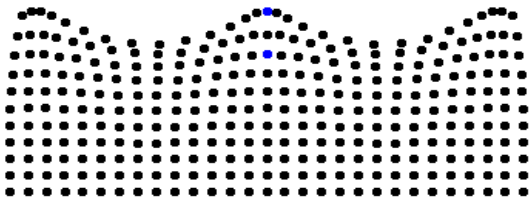
$$B = a \frac{\sinh(k(z+d))}{\sinh(kd)}$$

$$A_{SW} = a/kd$$

$$B_{SW} = a(1+z/d)$$

$$A_{DW} = B_{DW} = ae^{kz}$$

FACT - water particles do not move forward with water wave, but instead move in elliptical orbits. Waves are the forward motion of energy, not water!



WAVE PRESSURE

$$P = \gamma a \frac{\cosh(k(z+d))}{\cosh(kd)} \cos(kx - \sigma t) - \gamma z$$

Dynamic pressure
due to water particle motion

Hydrostatic

WAVE PRESSURE

$$P = \gamma a \frac{\cosh(k(z+d))}{\cosh(kd)} \cos(kx - \sigma t) - \gamma z$$

$$K_p = \frac{\cosh(k(z+d))}{\cosh(kd)} \quad \text{Pressure response factor}$$

$$\eta = a \cos(kx - \sigma t)$$

$$P = \gamma K_p \eta - \gamma z$$

WAVE PRESSURE

At $z=0$, $K_p=1$, $P=\gamma\eta$

$$\text{At } z=-d \quad P = \gamma\eta \frac{1}{\cosh(kd)} + \gamma d$$

DW ?

$$P=\gamma d$$

SW ?

$$P=\gamma(\eta+d)$$

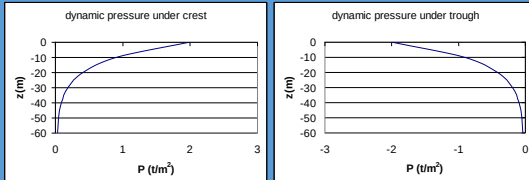
$$K_p = \frac{\cosh(k(z+d))}{\cosh(kd)}$$

$$P = \gamma K_p \eta - \gamma z$$

WAVE PRESSURE

At $z=0$, $K_p=1$, $P=\gamma\eta$

$$P = \gamma K_p \eta - \gamma z$$



Dyn wave pressure is max under wave crest
Min under wave trough

WAVE ENERGY

$$E = E_p + E_k$$

E_p is due to displacement of water surface
from the equilibrium level

E_k is due to water particle motion

$$E_T = \frac{1}{8} \gamma H^2 L$$

$$\bar{E} = E_T / L$$

Specific energy= total average wave energy
per unit surface area

$$\bar{E} = \frac{1}{8} \gamma H^2$$

WAVE POWER (ENERGY FLUX)

Energy transmission rate in the direction of wave propagation

$$P = \frac{\gamma H^2 L}{16T} \left(1 + \frac{2kd}{\sinh(2kd)} \right)$$

$$\bar{P} = \frac{1}{8} \rho g H^2 \frac{1}{2} \left(1 + \frac{4\pi d / L}{\sinh(4\pi d / L)} \right) \frac{L}{T}$$

$$P = \bar{E} C_n = \bar{E} C_g$$

WAVE POWER (ENERGY FLUX)

Small-amplitude water waves do not transmit mass as they propagate across a fluid, as the trajectories of the water particles are closed.

However, water waves do transmit energy.

The rate at which the energy is transferred is called the **energy Flux**, and for linear theory it is the rate at which work is being done by the fluid

work is being done by the dynamic pressure
 $Pd = (p + \rho g z)$ per unit width in the direction of wave propagation is

$$\mathcal{F} = \int_{-h}^n p_D \cdot u \, dz$$

WAVE POWER (ENERGY FLUX)

$$P = \frac{\gamma H^2 L}{16T} \left(1 + \frac{2kd}{\sinh(2kd)} \right)$$

$$\bar{P} = \frac{1}{8} \rho g H^2 \frac{1}{2} \left(1 + \frac{4\pi d / L}{\sinh(4\pi d / L)} \right) \frac{L}{T}$$

WAVE POWER (ENERGY FLUX)

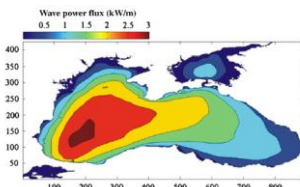
• For deep water

$$P = \frac{\rho g^2 H^2 T}{32\pi}$$

$$P = \frac{\rho g^2 H_s^2 T_e}{64\pi}$$

If the density of sea water for the Black Sea is taken as 1015 kg/m³,

$$P(\text{kilowatt/meter}) = 0.486 H_s^2 T_e$$

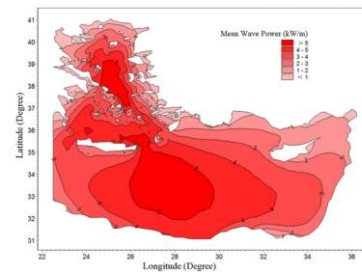


(Akpınar and Komurcu (2013))

Fig. 4. Spatial distribution of mean annual averaged wave power flux for all years based on 15-year SWAN hindcast results in the Black Sea (901 grid points between

the western part of the Black Sea (especially the south-west) is exposed to more energetic waves than the eastern part, and mean annual wave energy resource in the Black Sea is up to 3 kW/m. Besides, it was also determined that wave conditions have clearly seasonal dependency.

Mediterranean and Aegean Sea



Ayat B. (2013)

t the most energetic parts of the study area are central parts of the Eastern Mediterranean Sea and the Aegean Sea with mean wave power of about 5 kW/m based on 15-year hindcasted wave data set.

WAVE POWER (ENERGY FLUX)

$$P = \frac{\gamma H^2 L}{16T} \left(1 + \frac{2kd}{\sinh(2kd)} \right)$$

$$\bar{P} = \frac{1}{8} \rho g H^2 \frac{1}{2} \left(1 + \frac{4\pi d/L}{\sinh(4\pi d/L)} \right) \frac{L}{T}$$

$$P = \bar{E} C_n = \bar{E} C_g$$

Table 4.3 Gravity Wave Table (Lyngby, 1974)

GRAVITY WAVE TABLE									
$\frac{L}{d}$	$\frac{H}{L}$	$\frac{T}{L}$	$\frac{C_g}{L}$	$\frac{C_n}{L}$	$\frac{C_g}{C_n}$	$\frac{C_g}{C_n}$	$\frac{C_g}{C_n}$	$\frac{C_g}{C_n}$	$\frac{C_g}{C_n}$
1.00	0.000	0.000	0.000	0.000	1.000	1.000	1.000	1.000	1.000
1.05	0.001	0.001	0.001	0.001	0.999	0.999	0.999	0.999	0.999
1.10	0.002	0.002	0.002	0.002	0.998	0.998	0.998	0.998	0.998
1.15	0.003	0.003	0.003	0.003	0.997	0.997	0.997	0.997	0.997
1.20	0.004	0.004	0.004	0.004	0.996	0.996	0.996	0.996	0.996
1.25	0.005	0.005	0.005	0.005	0.995	0.995	0.995	0.995	0.995
1.30	0.006	0.006	0.006	0.006	0.994	0.994	0.994	0.994	0.994
1.35	0.007	0.007	0.007	0.007	0.993	0.993	0.993	0.993	0.993
1.40	0.008	0.008	0.008	0.008	0.992	0.992	0.992	0.992	0.992
1.45	0.009	0.009	0.009	0.009	0.991	0.991	0.991	0.991	0.991
1.50	0.010	0.010	0.010	0.010	0.990	0.990	0.990	0.990	0.990
1.55	0.011	0.011	0.011	0.011	0.989	0.989	0.989	0.989	0.989
1.60	0.012	0.012	0.012	0.012	0.988	0.988	0.988	0.988	0.988
1.65	0.013	0.013	0.013	0.013	0.987	0.987	0.987	0.987	0.987
1.70	0.014	0.014	0.014	0.014	0.986	0.986	0.986	0.986	0.986
1.75	0.015	0.015	0.015	0.015	0.985	0.985	0.985	0.985	0.985
1.80	0.016	0.016	0.016	0.016	0.984	0.984	0.984	0.984	0.984
1.85	0.017	0.017	0.017	0.017	0.983	0.983	0.983	0.983	0.983
1.90	0.018	0.018	0.018	0.018	0.982	0.982	0.982	0.982	0.982
1.95	0.019	0.019	0.019	0.019	0.981	0.981	0.981	0.981	0.981
2.00	0.020	0.020	0.020	0.020	0.980	0.980	0.980	0.980	0.980
2.05	0.021	0.021	0.021	0.021	0.979	0.979	0.979	0.979	0.979
2.10	0.022	0.022	0.022	0.022	0.978	0.978	0.978	0.978	0.978
2.15	0.023	0.023	0.023	0.023	0.977	0.977	0.977	0.977	0.977
2.20	0.024	0.024	0.024	0.024	0.976	0.976	0.976	0.976	0.976
2.25	0.025	0.025	0.025	0.025	0.975	0.975	0.975	0.975	0.975
2.30	0.026	0.026	0.026	0.026	0.974	0.974	0.974	0.974	0.974
2.35	0.027	0.027	0.027	0.027	0.973	0.973	0.973	0.973	0.973
2.40	0.028	0.028	0.028	0.028	0.972	0.972	0.972	0.972	0.972
2.45	0.029	0.029	0.029	0.029	0.971	0.971	0.971	0.971	0.971
2.50	0.030	0.030	0.030	0.030	0.970	0.970	0.970	0.970	0.970
2.55	0.031	0.031	0.031	0.031	0.969	0.969	0.969	0.969	0.969
2.60	0.032	0.032	0.032	0.032	0.968	0.968	0.968	0.968	0.968
2.65	0.033	0.033	0.033	0.033	0.967	0.967	0.967	0.967	0.967
2.70	0.034	0.034	0.034	0.034	0.966	0.966	0.966	0.966	0.966
2.75	0.035	0.035	0.035	0.035	0.965	0.965	0.965	0.965	0.965
2.80	0.036	0.036	0.036	0.036	0.964	0.964	0.964	0.964	0.964
2.85	0.037	0.037	0.037	0.037	0.963	0.963	0.963	0.963	0.963
2.90	0.038	0.038	0.038	0.038	0.962	0.962	0.962	0.962	0.962
2.95	0.039	0.039	0.039	0.039	0.961	0.961	0.961	0.961	0.961
3.00	0.040	0.040	0.040	0.040	0.960	0.960	0.960	0.960	0.960
3.05	0.041	0.041	0.041	0.041	0.959	0.959	0.959	0.959	0.959
3.10	0.042	0.042	0.042	0.042	0.958	0.958	0.958	0.958	0.958
3.15	0.043	0.043	0.043	0.043	0.957	0.957	0.957	0.957	0.957
3.20	0.044	0.044	0.044	0.044	0.956	0.956	0.956	0.956	0.956
3.25	0.045	0.045	0.045	0.045	0.955	0.955	0.955	0.955	0.955
3.30	0.046	0.046	0.046	0.046	0.954	0.954	0.954	0.954	0.954
3.35	0.047	0.047	0.047	0.047	0.953	0.953	0.953	0.953	0.953
3.40	0.048	0.048	0.048	0.048	0.952	0.952	0.952	0.952	0.952
3.45	0.049	0.049	0.049	0.049	0.951	0.951	0.951	0.951	0.951
3.50	0.050	0.050	0.050	0.050	0.950	0.950	0.950	0.950	0.950

Summary of Linear Wave Theory

RELATIVE DEPTH	SHALLOW WATER $\frac{L}{d} > \frac{1}{2}$	TRANSITIONAL WATER $\frac{1}{10} < \frac{L}{d} < \frac{1}{2}$	DEEP WATER $\frac{L}{d} < \frac{1}{10}$
1. Wave profile	Some Δz $\eta = \frac{H}{2} \cos \left(\frac{2\pi x}{L} - \frac{2\pi t}{T} \right) = \frac{H}{2} \cos \theta$	Some Δz $\eta = \frac{H}{2} \cos \left(\frac{2\pi x}{L} - \frac{2\pi t}{T} \right) = \frac{H}{2} \cos \theta$	Some Δz $\eta = \frac{H}{2} \cos \left(\frac{2\pi x}{L} - \frac{2\pi t}{T} \right) = \frac{H}{2} \cos \theta$
2. Wave celerity	$C = \frac{1}{2} \sqrt{\frac{gT}{\pi}}$	$C = \frac{1}{2} \sqrt{\frac{gT}{\pi}} \tanh \left(\frac{2\pi d}{L} \right)$	$C = C_0 \left(1 + \frac{3}{8} \frac{H^2}{L^2} \right)$
3. Wavelength	$L = T \sqrt{\frac{g}{k}} = CT$	$L = \frac{gT^2}{2\pi} \tanh \left(\frac{2\pi d}{L} \right)$	$L = L_0 \left(1 + \frac{3}{8} \frac{H^2}{L_0^2} \right) = C_0 T$
4. Group velocity	$C_g = C \left(1 + \frac{3}{8} \frac{H^2}{L^2} \right)$	$C_g = C \left(1 + \frac{3}{8} \frac{H^2}{L^2} \right) \tanh^2 \left(\frac{2\pi d}{L} \right)$	$C_g = \frac{1}{2} C \left(1 + \frac{3}{8} \frac{H^2}{L^2} \right)$
5. Water Particle Velocity	$u = \frac{H}{2} \sqrt{\frac{g}{k}} \cos \theta$	$u = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\cosh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \cos \theta$	$u = \frac{\pi H}{T} \cos \theta$
(a) Horizontal	$w = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$	$w = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$w = \frac{\pi H}{T} \sin \theta$
(b) Vertical	$u = \frac{H}{2} \sqrt{\frac{g}{k}} \cos \theta$	$u = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\cosh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \cos \theta$	$u = 2H \left(\frac{\pi}{T} \right)^2 \sin \theta$
6. Water Particle Accelerations	$a_x = \frac{\pi H}{T} \sqrt{\frac{g}{k}} \sin \theta$	$a_x = \frac{\pi H}{T} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$a_x = 2H \left(\frac{\pi}{T} \right)^2 \sin \theta$
(a) Horizontal	$a_z = -2H \left(\frac{\pi}{T} \right)^2 \cos \theta$	$a_z = -\frac{\pi H}{T} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \cos \theta$	$a_z = -2H \left(\frac{\pi}{T} \right)^2 \cos \theta$
(b) Vertical	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$
7. Water Particle Displacements	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$
(a) Horizontal	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$
(b) Vertical	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \frac{\sinh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} \sin \theta$	$\xi = \frac{H}{2} \sqrt{\frac{g}{k}} \sin \theta$
8. Subsurface Pressure	$p = p_0 (\eta - z)$	$p = p_0 \eta - \frac{\rho g H^2}{2} \frac{\cosh \left(\frac{2\pi (z+d)}{L} \right)}{\cosh \left(\frac{2\pi d}{L} \right)} - p_0 z$	$p = p_0 \eta - \frac{\rho g H^2}{2} - p_0 z$